

MIE 312 Tutorial 2 Summary

Ch.1 – Surface Tension & Viscometer

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Learning Objectives

- Understand the two surface effects of a liquid:
 - Surface tension Υ
 - Contact angle θ
- Calculate surface tension of droplets and small objects on water.
- Understand how viscometers measure viscosity.

Surface Tension

Basic Concept

- Not to be confused with buoyancy (floating is density based). Buoyancy is covered in Tutorial 4.
- Coefficient of surface tension Υ (pronounced “upsilon”):

Unit: N/m or lbf/ft

- Cut length dL :

$$F_{\text{surface tension}} = dL \times \Upsilon$$

- The force acts tangent to the edge of the liquid.
- Think of surface tension as a “cheap” plastic bag, trying to hold onto its dear life.

Water Droplet

- To calculate the surface tension of a water droplet, we can “pretend to” cut the droplet in half.
- For the droplet to not expand or contract, the forces must balance (gravity ignored).
- Pressure force and surface tension force:

$$F_{\text{Pressure}} = \Delta p \times \text{Area}, \quad F_{\text{Pressure}} = \pi R^2 \Delta p$$

$$F_{\text{Surface Tension}} = \text{Perimeter} \times \Upsilon, \quad F_{\text{Surf}} = 2\pi R \Upsilon$$

- Pressure difference:

$$\Delta P = P_{\text{inside}} - P_{\text{outside}}$$

e.g. $\Delta P = P_{\text{water droplet}} - P_{\text{air}}$

More Shapes (Fig. 1.11)

Pressure change across a curved interface due to surface tension:

- Interior of a liquid cylinder:

$$\Delta p = \frac{\Upsilon}{R}$$

- Interior of a spherical droplet:

$$\Delta p = \frac{2\Upsilon}{R}$$

- General curved interface (two principal radii R_1, R_2):

$$\Delta p = \Upsilon (R_1^{-1} + R_2^{-1})$$

- Note: for the liquid-cylinder case, we ignore the two end caps of the cylinder, since the surface tension acting on those ends points in a different direction than the pressure force.

Contact Angle θ

- Contact angle θ describes whether a liquid is “wetting” a surface.
 - Wet: $\theta < 90^\circ$
 - Nonwetting: $\theta > 90^\circ$
- Surface tension coefficient Υ (between two media), Eq. (1.29):

$$\Upsilon = \begin{cases} 0.0050 \text{ lbf/ft} = 0.073 \text{ N/m} & \text{air-water} \\ 0.033 \text{ lbf/ft} = 0.48 \text{ N/m} & \text{air-mercury} \end{cases}$$

- Examples: water on clean glass, $\theta \approx 0^\circ$ (wet); mercury on clean glass, $\theta = 130^\circ$ (nonwetting).

Vertical (Capillary) Tubes

- Imagine drawing a line from the edge of the liquid meniscus through the tube to determine the contact angle.
- Wetting liquid ($\theta < 90^\circ$): liquid rises in the tube, meniscus curves upward (concave).
- Non-wetting liquid ($\theta > 90^\circ$): liquid is depressed in the tube, meniscus curves downward (convex).

Example 1.8 — Capillary Rise

Derive an expression for the change in height h in a circular tube of a liquid with surface tension Υ and contact angle θ .

- Pressure force: $F_{\text{Pressure}} = \text{Pressure} \times \text{Area}$
- Apply Newton’s 2nd law in the vertical direction. The vertical component of the ring surface-tension force at the interface must balance the weight of the fluid column of height h :

$$F_{\text{surface tension}} \cos \theta = W$$

$$2\pi R \Upsilon \cos \theta = \gamma \pi R^2 h$$

- Solving for h :

$$h = \frac{2\Upsilon \cos \theta}{\rho g R}$$

where specific weight $\gamma = \rho g$.

- The capillary height increases inversely with tube radius R , and is positive (rise) if $\theta < 90^\circ$ (wetting), negative (capillary depression) if $\theta > 90^\circ$.

Numerical check ($R = 1$ mm):

- Water–air–glass: $\theta \approx 0^\circ$, $\Upsilon = 0.073$ N/m, $\rho = 1000$ kg/m³:

$$h = \frac{2(0.073) \cos 0^\circ}{(1000)(9.81)(0.001)} = 0.015 \text{ m} = 1.5 \text{ cm}$$

- Mercury–air–glass: $\theta = 130^\circ$, $\Upsilon = 0.48$ N/m, $\rho = 13,600$ kg/m³:

$$h = \frac{2(0.48) \cos 130^\circ}{(13,600)(9.81)(0.001)} = -0.0046 \text{ m} = -0.46 \text{ cm}$$

Capillary Viscometer

- Another way of measuring viscosity.
 1. Testing liquid is first stored in the lower chamber.
 2. Liquid is then suction-pumped to the upper timing marker.
 3. Suction is released, allowing the liquid to flow back down.
 4. Time taken for the liquid to travel from the upper to the lower timing mark is measured.
- Viscosity of the liquid is then calculated from the flow time.
- Longer time = more viscous.

Rotating Viscometer

- Used to measure the viscosity of fluids.
- Shear stress τ is generated by the rotating spindle. Recall:

$$\tau = \mu \frac{du}{dy}$$

- Torque from the shear stress is measured by the motor.
- More torque = more viscous.
- Rotating viscometers come in different shapes (e.g. a straight cylindrical spindle, or a cone-and-plate spindle as in Problem 3 below).